جامعة بابل

كلية العلوم للبنات

قسم الحاسوب

الاسم:سجى حيدر

الموضوع: نظام ادارة مكتبة



عمل نظام ادارة المكتبة

تبدأ بوصول العضو الى المكتبة حيث يتم الاستفسار عن معلومات الكتاب (رقم الكتاب,عنوان الكتاب,المؤلف,السعر,التوفر) ويتم الاستفسار عن المعلومات الاخرى مثلا عن الناشر او اذا كان الكتاب مستعارا ام لا .

يمكن للعضو استعارة العديد من الكتب، ويمكن تصفح الكتاب من قبل العديد من الأعضاء. يمكن للناشر نشر العديد من الكتب؛ ومع ذلك، يجب أن يتم نشر الكتاب من قبل ناشر واحد.

يجب أن تسجل قاعدة البيانات تاريخ الاستعارة عند قيام العضو باستعارة أي كتاب.

يتضمن نظام إدارة المكتبات عادة قاعدة بيانات مترابطة وبرمجية للتفاعل مع قاعدة البيانات تلك، وواجهتي استخدام (رسوميات عادة)، إحداهما للمستخدمين وأخرى لطاقم عمل المكتبة. معظم برمجيات أنظمة إدارة المكتبات تقسم الوظائف إلى برامج منفصلة أو وحدات، تتكامل معا في واجهة استخدام موحدة. أمثلة تلك الوحدات قد تتضمن:

الاستحواذ (التزويد) (طلب المواد، إستقبالها وفوترتها)
الفهرسة (تصنيف وتبويب وفهرسة المواد)
التداول والاعارة (إعارة المواد للمستعيرين وإسترجاعها)

يتكون النظام من 4 جداول

كيان"BOOK" :الجدول الاول يكون عن الكتب في المكتبة.

book_ID: معرف الكتاب في المكتبة.

Author: مؤلف الكتاب.

Title: عنوان الكتاب.

Price:سعر الكتاب

Availability:توفر الكتاب.

كيان"Member":الجدول الثاني يكون عبارة عن العضو .

member_ID:معرف العضو .

member_type:نوع العضو (ذكر ,انثى).

member_name :اسم العضو.

كيان "Publisher":الجدول الثالث يكون عبارة عن الناشر

publisher_ID:معرف الناشر.

publisher_name:اسم الناشر.

publisher_address:عنوان الناشر.

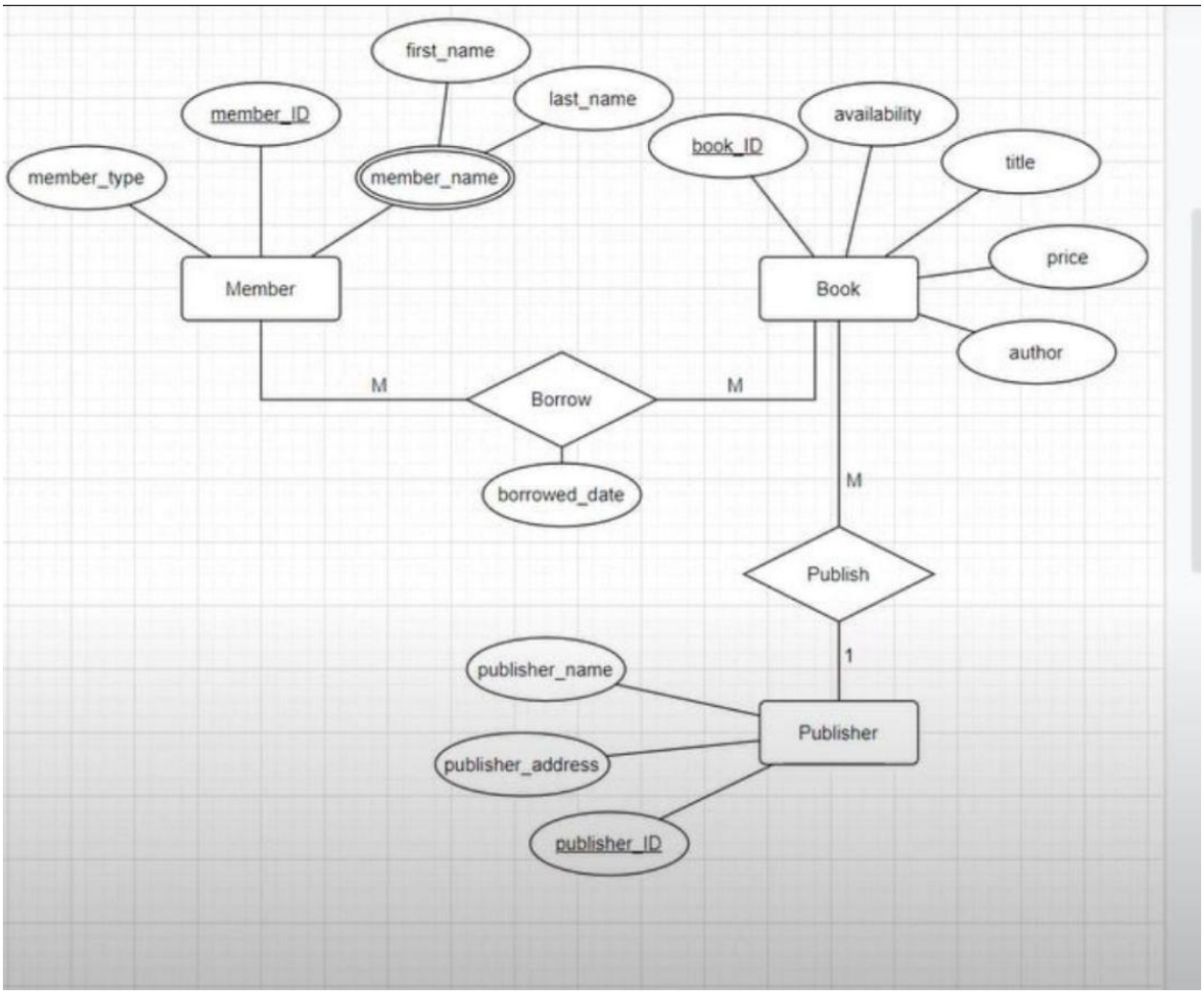
كيان " Borrowed Book":الجدول الرابع يكون عن استعارة

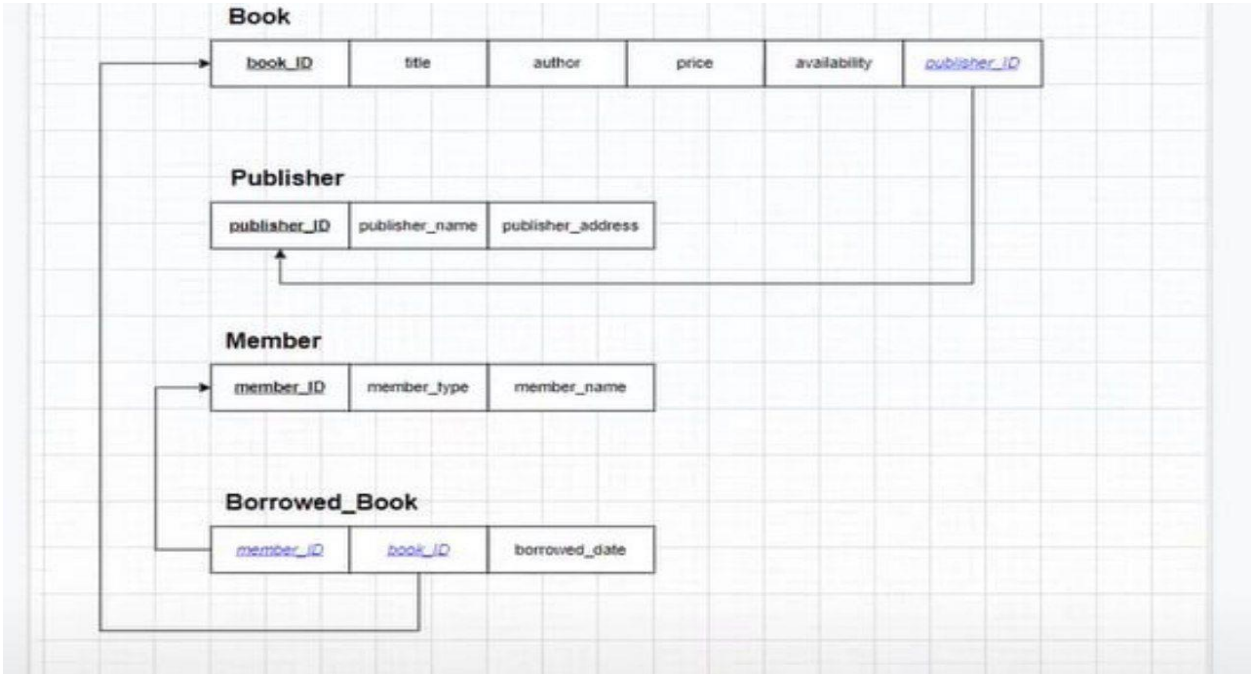
Member_ID:معرف العضو.

book _ID:معرف الكتاب.

borrowed_date:تاريخ الاقتراض.

fine:غرامة .


first_name
last_name
member_ID
member_type
member_name
book_ID
availability
title
price
author
Member
Book
M
Borrow
M
borrowed_date
M
Publish
1
publisher_name
Publisher
publisher_address
publisher_ID


Book
book_ID
title
author
price
availability
publisher_ID
Publisher
publisher_ID
publisher_name
publisher_address
Member
member_ID
member_type
member_name
Borrowed_Book
member_ID
book_ID
borrowed_date

كود الجداول في Sql

```
use library;
go
CREATE TABLE Book (
book_ID int PRIMARY KEY,
title VARCHAR(255),
author VARCHAR(255),
price DECIMAL(10),
availability VARCHAR(10),
publisher_ID INT,
FOREIGN KEY (publisher_ID) REFERENCES Publisher(publisher_ID),
);
Go


CREATE TABLE Member (
member_ID INT PRIMARY KEY,
member_type VARCHAR(255),
member_name VARCHAR(255),
);
Go

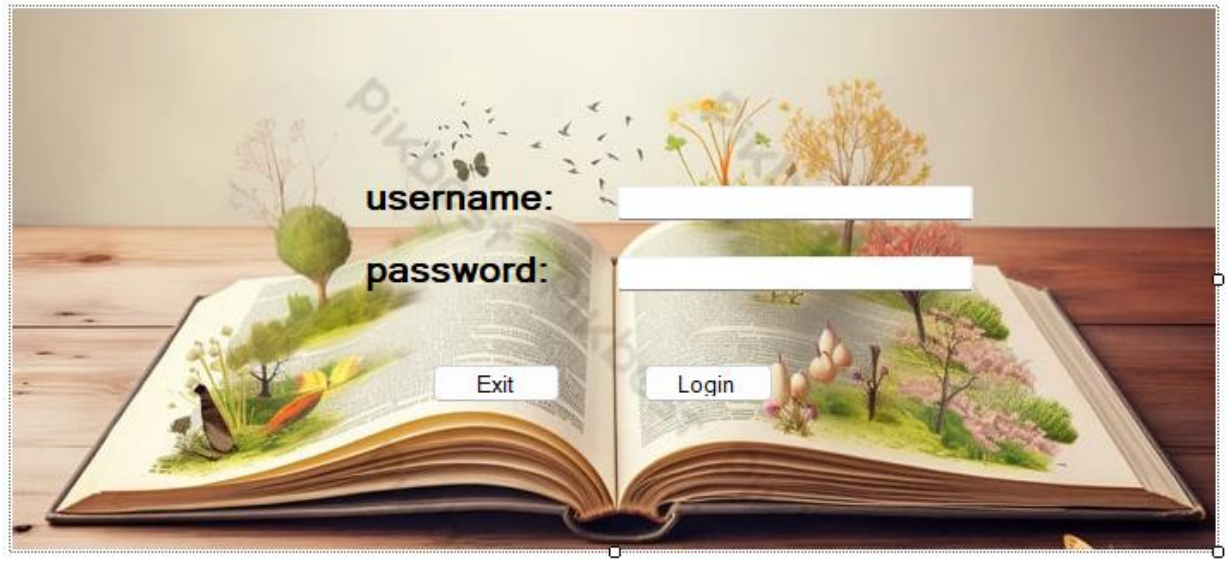

CREATE TABLE Publisher (
publisher_ID INT PRIMARY KEY,
publisher_name VARCHAR(255),
publisher_address VARCHAR(255),
 );
 Go




CREATE TABLE Borrowed_Book (
member_ID INT,
book_ID INT,
borrowed_date DATE,
fine DECIMAL(10,2),
PRIMARY KEY (member_ID, book_ID),
FOREIGN KEY (member_ID) REFERENCES Member(member_ID),
FOREIGN KEY (book_ID) REFERENCES Book(book_ID),
);
Go
```

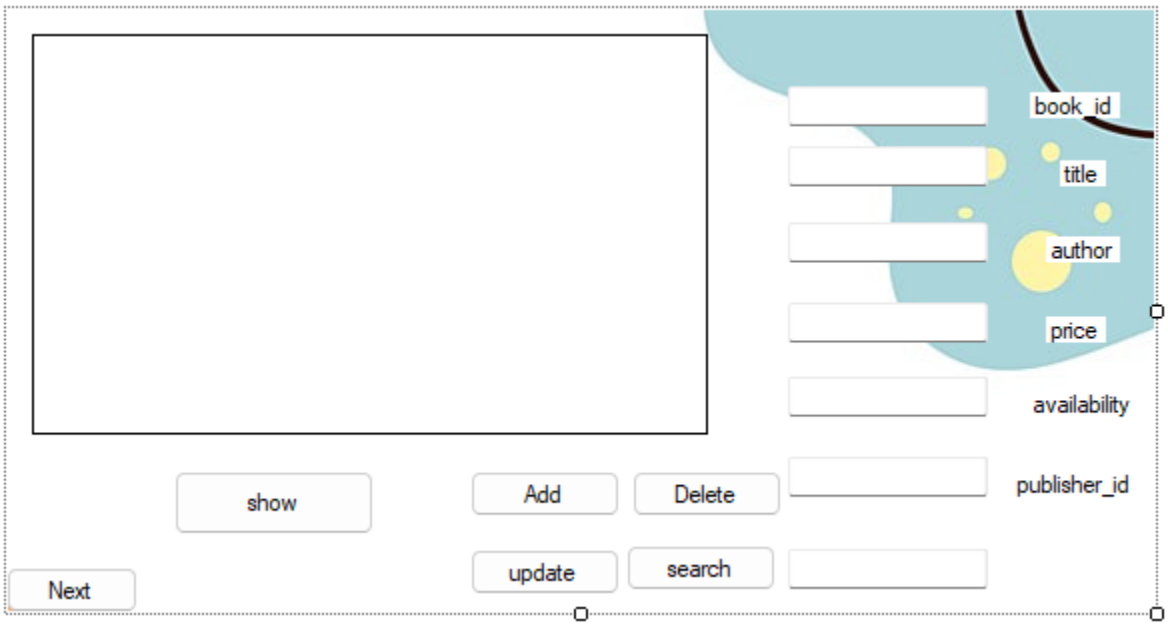
واجهات النظام

الواجهة الاولى للدخول الى النظام




```
Public Class Form1
Private Sub Button1_Click(sender As System.Object, e As
System.EventArgs) Handles Button1.Click
        Hide()
        Form2.Show()
    End Sub
```

الواجهة الثانية للنظام




```
Imports System.Data
Imports System.Data.SqlClient
Public Class Form2
    Dim conn As New SqlConnection("Data Source=.\sqlexpress;Initial
Catalog=library;Integrated Security=True")
    Dim ds As New DataSet
    Dim apdeter As SqlDataAdapter
Private Sub Button1_Click(sender As System.Object, e As System.EventArgs)
Handles Button1.Click
        apdeter = New SqlDataAdapter("select * from Book", conn)
        apdeter.Fill(ds, "book1")
        DataGridView1.DataSource = ds.Tables("book1")
    End Sub
Private Sub Button2_Click(sender As System.Object, e As System.EventArgs)
Handles Button2.Click
        Dim cmd As New SqlCommand("insert into Book
(book_ID,title,author,price,availability,publisher_ID) values(' " &
TextBox1.Text & " ',' " & TextBox2.Text & " ',' " & TextBox3.Text & " ', " &
TextBox4.Text & " ',' " & TextBox5.Text & " ',' " & TextBox6.Text & " ')",
conn)
        conn.Open()
        cmd.ExecuteNonQuery()
        conn.Close()
        MsgBox("its ok")
        TextBox1.Clear()
        TextBox2.Clear()
        TextBox3.Clear()
```

```
        TextBox4.Clear()
        TextBox5.Clear()
        TextBox6.Clear()

    End Sub
Private Sub Button3_Click(sender As System.Object, e As System.EventArgs)
Handles Button3.Click
        Dim cmd As New SqlCommand("update Book set author='" & TextBox3.Text
& "'where author=ali", conn)
        conn.Open()
        cmd.ExecuteNonQuery()
        conn.Close()
        MsgBox("update")
        TextBox1.Clear()
        TextBox2.Clear()
        TextBox3.Clear()
        TextBox4.Clear()
        TextBox5.Clear()
        TextBox6.Clear()
    End Sub

Private Sub Button4_Click(sender As System.Object, e As System.EventArgs)
Handles Button4.Click
        Dim cmd As New SqlCommand("delete from Book where author=' " &
TextBox3.Text & "'", conn)
        conn.Open()
        cmd.ExecuteNonQuery()
        conn.Close()
        MsgBox("its ok")
    End Sub
Private Sub Button5_Click(sender As System.Object, e As System.EventArgs)
Handles Button5.Click
        Dim ds1 As New DataTable
        apdeter = New SqlDataAdapter("select * from Book where author='" &
TextBox7.Text & "' ", conn)
        apdeter.Fill(ds1)
        If (ds1.Rows.Count > 0) Then
            DataGridView1.DataSource = ds1
        Else
            MsgBox("no data found")
        End If
    End Sub
Private Sub Button6_Click_1(sender As System.Object, e As System.EventArgs)
Handles Button6.Click
        Hide()
        Form3.Show()
    End Sub
End Class
```

الواجهة الثالثة للنظام




```
Imports System.Data
Imports System.Data.SqlClient
Public Class Form3
    Dim conn As New SqlConnection("Data Source=.\sqlexpress;Initial
Catalog=library;Integrated Security=True")
    Dim ds As New DataSet
    Dim apdeter As SqlDataAdapter
    Private Sub Button1_Click(sender As System.Object, e As
System.EventArgs) Handles Button1.Click
        apdeter = New SqlDataAdapter("select * from Member", conn)
        apdeter.Fill(ds, " Member1")
        DataGridView1.DataSource = ds.Tables(" Member1")
    End Sub

    Private Sub Button2_Click(sender As System.Object, e As
System.EventArgs) Handles Button2.Click
        Dim cmd As New SqlCommand("insert into Member (
memberk_ID,member_type,member_name) values(' " & TextBox1.Text & " ','
" & TextBox2.Text & " ',' " & TextBox3.Text & " ')", conn)
        conn.Open()
        cmd.ExecuteNonQuery()
        conn.Close()
        MsgBox("its ok")
        TextBox1.Clear()
        TextBox2.Clear()
        TextBox3.Clear()
    End Sub

    Private Sub Button3_Click(sender As System.Object, e As
System.EventArgs) Handles Button3.Click
```

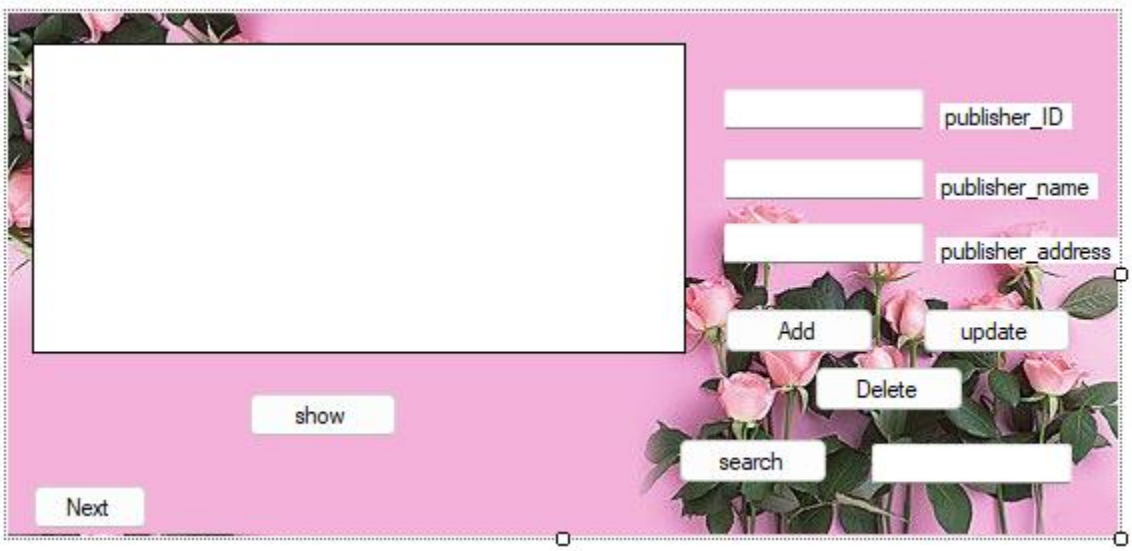
```
        Dim cmd As New SqlCommand("update  Member set member_name='" &
TextBox3.Text & "'where member_name=ali", conn)
        conn.Open()
        cmd.ExecuteNonQuery()
        conn.Close()
        MsgBox("update")
        TextBox1.Clear()
        TextBox2.Clear()
        TextBox3.Clear()
    End Sub

    Private Sub Button4_Click(sender As System.Object, e As
System.EventArgs) Handles Button4.Click
        Dim cmd As New SqlCommand("delete from Member where
member_name=' " & TextBox3.Text & "'", conn)
        conn.Open()
        cmd.ExecuteNonQuery()
        conn.Close()
        MsgBox("its ok")
    End Sub

    Private Sub Button5_Click(sender As System.Object, e As
System.EventArgs) Handles Button5.Click
        Dim ds1 As New DataTable
        apdeter = New SqlDataAdapter("select * from Member where
member_name='" & TextBox4.Text & "' ", conn)
        apdeter.Fill(ds1)
        If (ds1.Rows.Count > 0) Then
            DataGridView1.DataSource = ds1
        Else
            MsgBox("no data found")
        End If
    End Sub

    Private Sub Button6_Click(sender As System.Object, e As
System.EventArgs) Handles Button6.Click
        Hide()
        Form4.Show()
    End Sub
```

الواجهة الرابعة للنظام




```
Imports System.Data
Imports System.Data.SqlClient
Public Class Form4
    Dim conn As New SqlConnection("Data Source=.\sqlexpress;Initial
Catalog=library;Integrated Security=True")
    Dim ds As New DataSet
    Dim apdeter As SqlDataAdapter

    Private Sub Button1_Click(sender As System.Object, e As
System.EventArgs) Handles Button1.Click
        apdeter = New SqlDataAdapter("select * from Publisher", conn)
        apdeter.Fill(ds, " Publisher1")
        DataGridView1.DataSource = ds.Tables(" Publisher1")
    End Sub

    Private Sub Button2_Click(sender As System.Object, e As
System.EventArgs) Handles Button2.Click
        Dim cmd As New SqlCommand("insert into Publisher(
publisher_ID,publisher_name,publisher_address) values(' " &
TextBox1.Text & " ',' " & TextBox2.Text & " ',' " & TextBox3.Text & "
')", conn)
        conn.Open()
        cmd.ExecuteNonQuery()
        conn.Close()
        MsgBox("its ok")
        TextBox1.Clear()
        TextBox2.Clear()
        TextBox3.Clear()
```

```
    End Sub

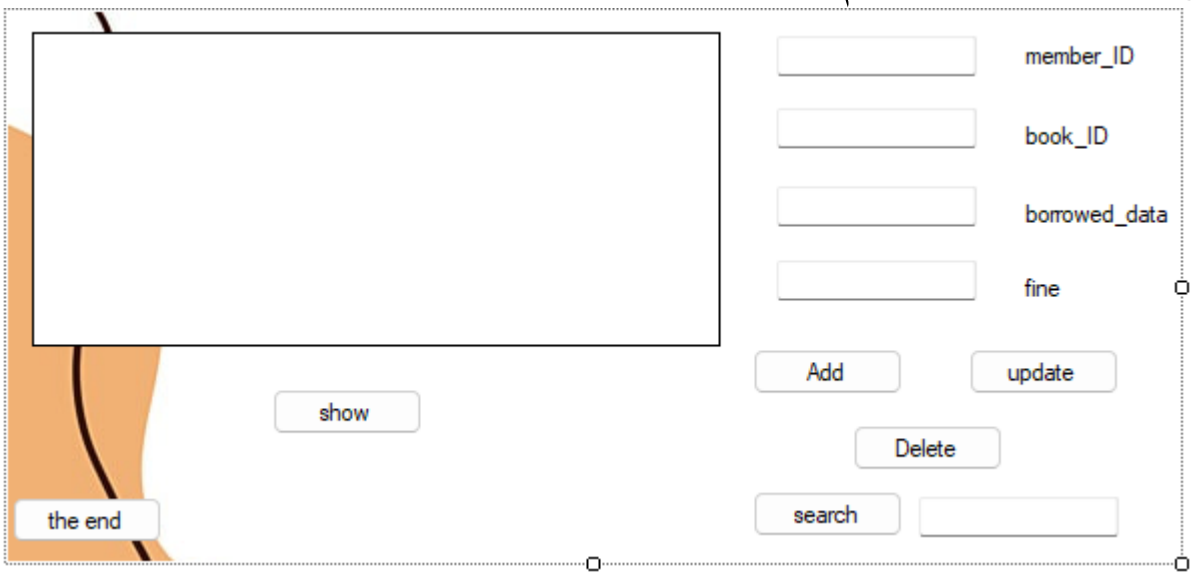
    Private Sub Button3_Click(sender As System.Object, e As
System.EventArgs) Handles Button3.Click
        Dim cmd As New SqlCommand("update  Publisher set
publisher_name='" & TextBox2.Text & "'where publisher_name=ali", conn)
        conn.Open()
        cmd.ExecuteNonQuery()
        conn.Close()
        MsgBox("update")
        TextBox1.Clear()
        TextBox2.Clear()
        TextBox3.Clear()
    End Sub

    Private Sub Button4_Click(sender As System.Object, e As
System.EventArgs) Handles Button4.Click
        Dim cmd As New SqlCommand("delete from Publisher where
publisher_name=' " & TextBox2.Text & "'", conn)
        conn.Open()
        cmd.ExecuteNonQuery()
        conn.Close()
        MsgBox("its ok")
    End Sub

    Private Sub Button5_Click(sender As System.Object, e As
System.EventArgs) Handles Button5.Click
        Dim ds1 As New DataTable
        apdeter = New SqlDataAdapter("select * from Publisher where
publisher_name='" & TextBox4.Text & "' ", conn)
        apdeter.Fill(ds1)
        If (ds1.Rows.Count > 0) Then
            DataGridView1.DataSource = ds1
        Else
            MsgBox("no data found")
        End If
    End Sub

    Private Sub Button6_Click(sender As System.Object, e As
System.EventArgs) Handles Button6.Click
        Hide()
        Form5.Show()
    End Sub
End Class
```

الواجهة الخامسة للنظام




```
Imports System.Data
Imports System.Data.SqlClient
Public Class Form5
    Dim conn As New SqlConnection("Data Source=.\sqlexpress;Initial
Catalog=library;Integrated Security=True")
    Dim ds As New DataSet
    Dim apdeter As SqlDataAdapter

    Private Sub Button1_Click(sender As System.Object, e As System.EventArgs)
Handles Button1.Click
        apdeter = New SqlDataAdapter("select * from Borrowed_Book", conn)
        apdeter.Fill(ds, " Borrowed_Book1")
        DataGridView1.DataSource = ds.Tables(" Borrowed_Book1")
    End Sub

    Private Sub Button2_Click(sender As System.Object, e As System.EventArgs)
Handles Button2.Click
        Dim cmd As New SqlCommand("insert into Borrowed_Book(
member_ID,book_ID,borrowed_data,fine) values(' " & TextBox1.Text & " ',' " &
TextBox2.Text & " ',' " & TextBox3.Text & " ',' " & TextBox4.Text & " ')",
conn)
        conn.Open()
        cmd.ExecuteNonQuery()
        conn.Close()
        MsgBox("its ok")
        TextBox1.Clear()
        TextBox2.Clear()
        TextBox3.Clear()
        TextBox4.Clear()
    End Sub
```

```
    Private Sub Button3_Click(sender As System.Object, e As System.EventArgs)
Handles Button3.Click
        Dim cmd As New SqlCommand("update  Borrowed_Book set borrowed_data='"
& TextBox3.Text & "'where borrowed_data=2024\4\4", conn)
        conn.Open()
        cmd.ExecuteNonQuery()
        conn.Close()
        MsgBox("update")
        TextBox1.Clear()
        TextBox2.Clear()
        TextBox3.Clear()
        TextBox4.Clear()
    End Sub

    Private Sub Button4_Click(sender As System.Object, e As System.EventArgs)
Handles Button4.Click
        Dim cmd As New SqlCommand("delete from Borrowed_Book where
borrowed_data=' " & TextBox3.Text & "'", conn)
        conn.Open()
        cmd.ExecuteNonQuery()
        conn.Close()
        MsgBox("its ok")
    End Sub

    Private Sub Button5_Click(sender As System.Object, e As System.EventArgs)
Handles Button5.Click
        Dim ds1 As New DataTable
        apdeter = New SqlDataAdapter("select * from Borrowed_Book  where
borrowed_data='" & TextBox5.Text & "' ", conn)
        apdeter.Fill(ds1)
        If (ds1.Rows.Count > 0) Then
            DataGridView1.DataSource = ds1
        Else
            MsgBox("no data found")
        End If
    End Sub
End Class
```